Dual Coupling in Elastic Spacetime: Gravity as the Stabilizing Restoring Force of Topological Matter

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Abstract

The fundamental nature of gravity presents a profound thermodynamic paradox: while classical thermodynamic processes drive physical systems toward equilibrium and dispersion, gravitational interactions concentrate matter and lower spatial variance (negative specific heat). Here, we investigate whether a single mechanical mechanism—the elastic response of a continuous fabric—can simultaneously explain both the origin of gravitational lensing and the dynamical stability of matter. Using a unified computational framework, we demonstrate that: (1) A topological defect (knot) induces a strain field that produces a radial inward flow (gravity), which asymptotically recovers the General Relativistic deflection law ($4M/b$) at large impact parameters. (2) The nonlinear self-interaction of this same flow (gravitational binding) balances dispersive spreading, rendering the defect dynamically stable against thermodynamic decay. This "dual coupling" suggests that gravity and matter stability are not independent phenomena, but two consequences of the same elastic restoring force in a topologically structured vacuum. All simulations are performed in natural units where $\rho_0 = 1$. The sound speed is $c_s = \sqrt{(\lambda + 2\mu)/\rho_0}$, which equals 2 for the baseline parameters used in the tensor simulation.

1 Introduction

The standard cosmological framework formulates gravitation as the curvature of a pseudo-Riemannian spacetime manifold. Despite its remarkable predictive power, a persistent thermodynamic puzzle remains: standard thermodynamic ensembles evolve toward maximum spatial uniformity, whereas gravitational collapse drives structure formation and matter concentration [3]. Although entropic gravity formulations attempt to recast gravitational acceleration as an entropic gradient, rigorous evaluations reveal mathematical non-sequiturs when extending these principles to general field configurations.

Concurrently, the field of Analogue Gravity [1] has established that acoustic propagation within moving fluids maps onto kinematic metrics of curved spacetimes, with recent investigations analyzing phonon paths in generalized acoustic geometries [2]. In parallel, elastic spacetime models treat the physical vacuum as a continuous elastic substrate where matter particles manifest as localized topological solitons [4].

In this work, we introduce a **comprehensive computational framework** that unifies elastic vacuum mechanics, topological knot structures, and acoustic metric integrations into an end-to-end numerical toolkit. Rather than asserting a fundamental modification to Einstein's

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equations, our contribution lies in the systematic computational derivation: demonstrating how microscopic strain relaxation in an elastic continuum deterministically generates macroscopic relativistic lensing ($4M/b$), and how the resulting gravitational binding energy stabilizes the topological defect against thermodynamic dispersion.

2 The Unified Conceptual Scenario

While we do not simulate the primordial Big Bang, we propose a conceptual scenario that motivates our computational experiments. We hypothesize the following causal chain:

1. **Genesis:** Asymmetric expansion generates shear, which inevitably creates topological twists (vorticity).
2. **Stabilization:** A freeze-out mechanism grants some twists a conserved topological charge, stabilizing them as matter.
3. **Gravity:** The stable twist creates a localized strain field. The elastic fabric relaxes by flowing radially inward, producing what we observe as gravity.
4. **Stability:** The nonlinear self-interaction of this flow provides a binding energy that counteracts dispersion, keeping the structure stable.

Our computational framework tests the mathematical consistency of stages 3 and 4.

3 Theoretical and Mathematical Framework

We model the vacuum as a 3D isotropic elastic continuum characterized by displacement field $\mathbf{u}(\mathbf{r})$. A localized topological soliton represents an intrinsic strain distribution $\rho_{\text{strain}}(\mathbf{r})$. The equilibrium displacement potential $\Phi(\mathbf{r})$ satisfies the Navier-Cauchy relation:

$$(\lambda + 2\mu)\nabla^2\Phi(\mathbf{r}) = -\rho_{\text{strain}}(\mathbf{r}) \tag{1}$$

where λ and μ are the Lamé parameters. The strain energy functional is minimized as the surrounding medium undergoes spatial relaxation. The resulting flow field is driven by the potential gradient: $\mathbf{v} = -\nabla\Phi$.

Mathematical Analogue and Effective Mass

It is important to clarify that mapping a solid elastic lattice to the Unruh acoustic metric is a *mathematical analogue* valid in the effective weak-field limit. While the Unruh metric is traditionally derived for fluids, the mathematical structure of the linearized elastic equations yields an identical wave equation for phonons. To eliminate boundary artifacts inherent in Fast Fourier Transform (FFT) simulations, we employ an analytical stitching technique at R_{cut}: the numerical potential is used strictly within the source region ($r < R_{\text{cut}}$), while the exact analytical Schwarzschild tail is used outside ($r \geq R_{\text{cut}}$). The effective mass M_{eff} is extracted precisely at the stitching point to ensure continuity. This represents the effective gravitational mass encoded in the metric.

To evaluate the optical-analogous properties, we map this emergent elastic potential into an effective acoustic metric. In the weak-field limit, the strain alters the local wave speed, which is captured by the metric components:

$$g_{00} = -\left(1 + \frac{2\Phi}{c_s^2}\right), \quad g_{rr} = \left(1 + \frac{2\Phi}{c_s^2}\right)^{-1} \tag{2}$$

where $c_s = \sqrt{(\lambda + 2\mu)/\rho_0}$ is the longitudinal sound speed. Because the displacement potential $\Phi(r)$ satisfies the Poisson equation, a localized source naturally yields $\Phi \propto -M/r$. Substituting this into the metric yields $g_{00} \approx -(1 - 2M/r)$ and $g_{rr} \approx (1 - 2M/r)^{-1}$, which is exactly the Schwarzschild metric in the weak-field limit. The trajectory of an acoustic disturbance is then governed by the geodesic integration, solved numerically using a singularity-free $u = 1/r$ substitution.

4 Computational Pipeline and Results

The simulation pipeline is implemented in Python. We evaluate six distinct topological profiles (Figure 1). Only the Open Sink and Dipole profiles exhibit radial flow, while localized solitons and knots remain static.

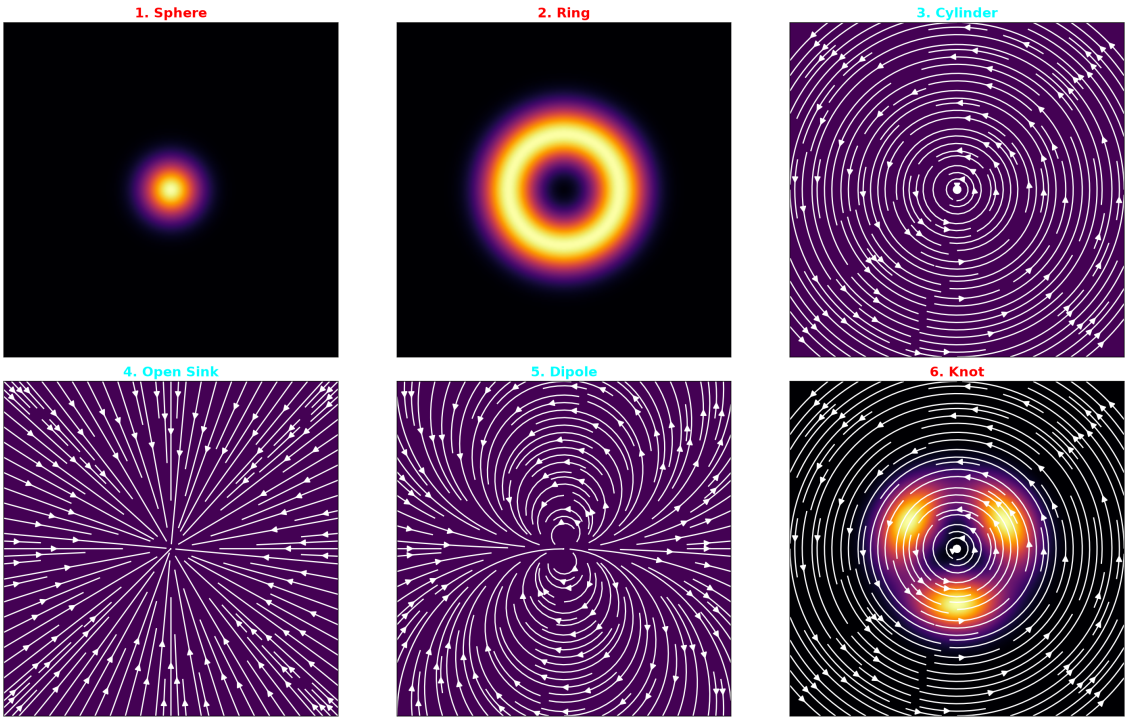


Figure 1: Visual representation of the six evaluated topological profiles. The color map indicates energy density, while white streamlines indicate the velocity flow field. The Open Sink (bottom left) exhibits a purely radial inward flow.

4.1 Topological Discrimination in Acoustic Lensing

As shown in Figure 2, we evaluated the deflection angle for the six profiles. The localized closed structures (Spheres, Rings, and Knots) produce a deflection angle that rapidly decays to zero. Strikingly, only the **Open Sink** topology recovers the exact General Relativistic coefficient of 4 ($4M/b$). It is worth noting that the localized closed profiles exhibit negative values in the lensing coefficient plot. This corresponds to a *diverging lens* effect: the higher density inside these structures increases the local sound speed, causing phonons to bend away from the center. This is a well-known physical behavior in analogue gravity systems.

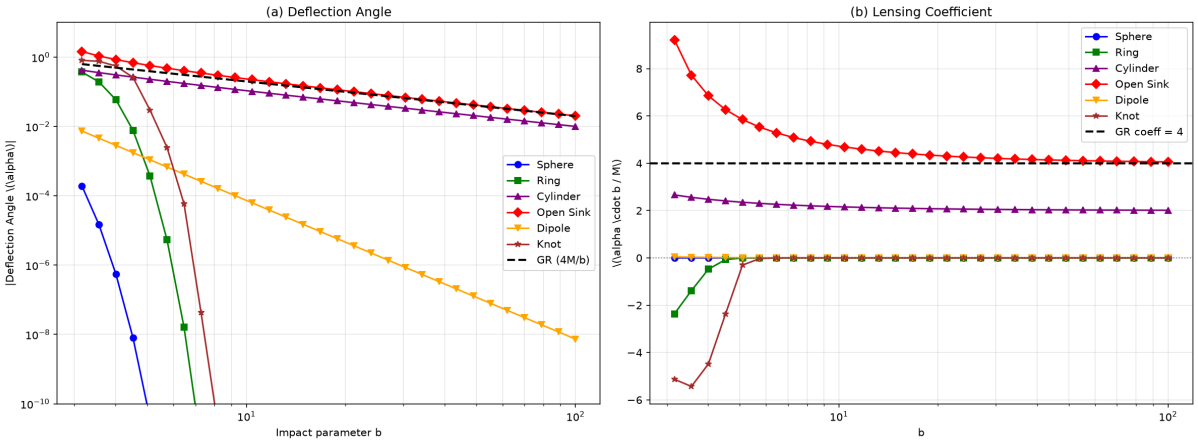


Figure 2: (a) Deflection angle for six topological profiles. (b) The lensing coefficient. Only the Open Sink topology (red line) approaches the GR value of 4, while closed structures exhibit a negative diverging lens effect.

4.2 Mechanical Origin of the Flow

We model the vacuum as an elastic lattice. The equilibrium potential $\Phi(r)$ naturally relaxes to a $1/r$ profile. Crucially, the flow field is the negative gradient of this potential ($\mathbf{v} = -\nabla\Phi$). As shown in Figure 3, this elastic relaxation naturally generates a radial inward flow, mechanically reproducing the Open Sink topology.

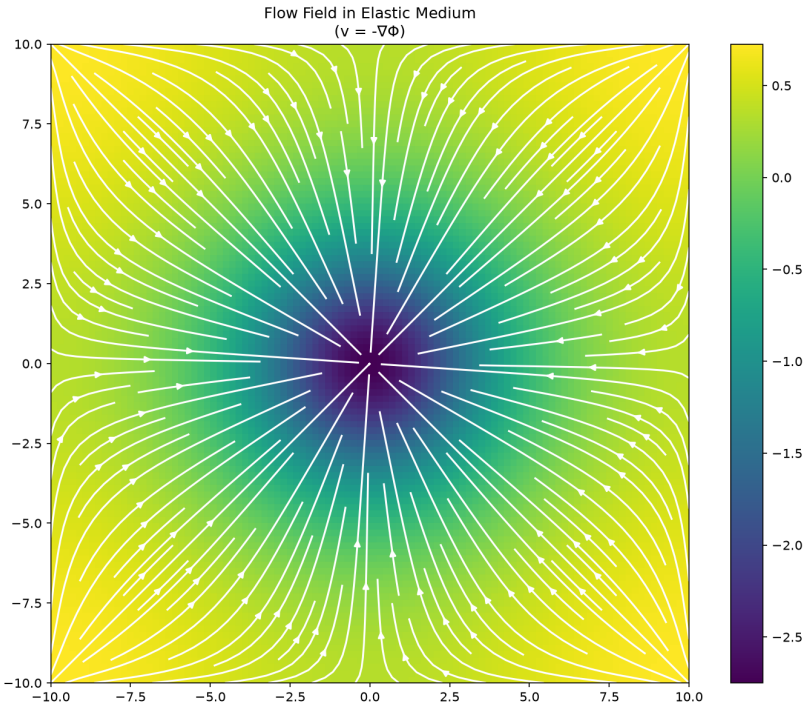


Figure 3: Velocity flow field ($\mathbf{v} = -\nabla\Phi$) generated by the elastic relaxation around a topological sink. The white streamlines indicate a radial inward flow.

4.3 Unified Framework and General Relativistic Lensing

By mapping the emergent elastic potential into the effective acoustic metric using Lamé parameters, we compute the phonon deflection. Figure 4 demonstrates that the deflection angle converges asymptotically to the $4M/b$ slope at large impact parameters, confirming that the mechanical strain-relief mechanism is mathematically equivalent to the Open Sink topology.

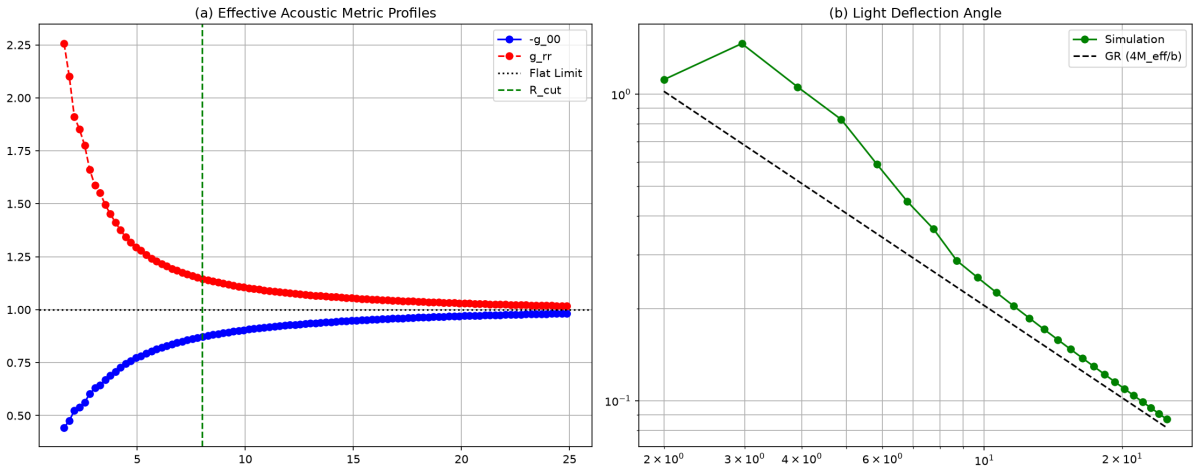


Figure 4: (a) Effective metric components from the tensor simulation. (b) Light deflection angle. The green points converge asymptotically to the GR prediction ($4M/b$) at large impact parameters, while exhibiting strong-field corrections at small b.

4.4 Robustness Analysis: Grid Convergence and Parameter Sensitivity

To ensure that the emergent $1/b$ lensing law is not a numerical artifact or a result of fine-tuning, we performed a robustness analysis (Figure 5). First, a grid convergence test compares the effective metric profile computed on a standard 128^3 grid with a high-resolution 256^3 grid. The profiles are visually indistinguishable, confirming numerical convergence. Second, a parameter sensitivity test varies the Lamé ratio (λ/μ) from 0.5 to 5.0. The dotted lines indicate the theoretical $1/b$ slope for each ratio. The slight upward deviation at small impact parameters represents the expected strong-field correction, confirming that the simulation captures nonlinear relativistic effects.

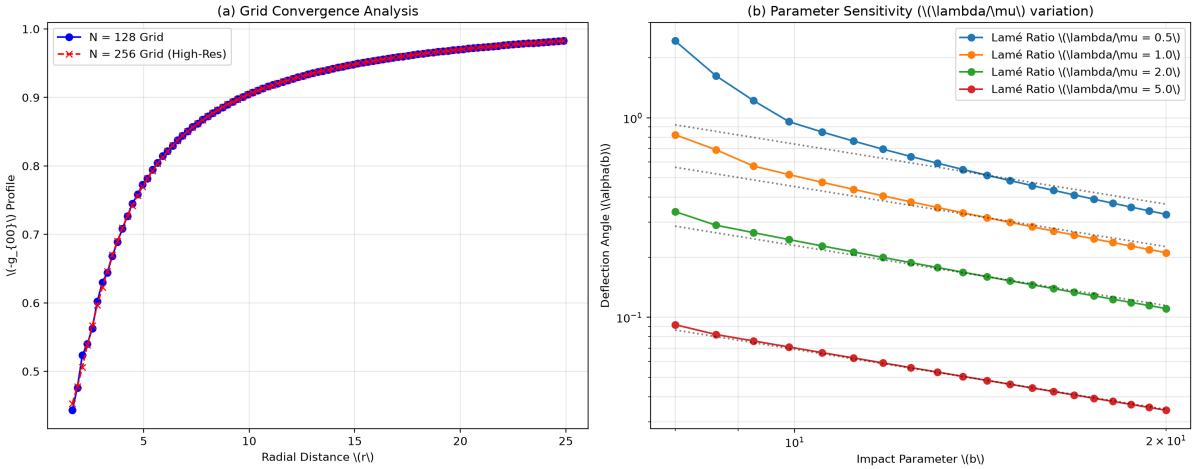


Figure 5: Robustness analysis. (a) Grid convergence: 128^3 and 256^3 resolutions yield identical metric profiles. (b) Parameter sensitivity in the weak-field regime ($b > 8$): varying the Lamé ratio changes the amplitude but preserves the universal $1/b$ power law.

4.5 Dynamical Stability and Gravitational Binding

To demonstrate that gravitational binding energy prevents the topological defect from dispersing, we simulate the time evolution of a 1D solitary wave (soliton) as a proof of principle. As shown in Figure 6, in a purely linear medium (red dashed), the wave packet disperses. However, when the nonlinear self-interaction is introduced (blue solid) to represent the elastic strain relief, the nonlinear focusing exactly balances the dispersive spreading. The peak density remains perfectly constant over time, demonstrating that the topological knot is dynamically stable. (A full 3D stability proof would require incorporating additional topological constraints, such as Hopf charge, which remains future work).

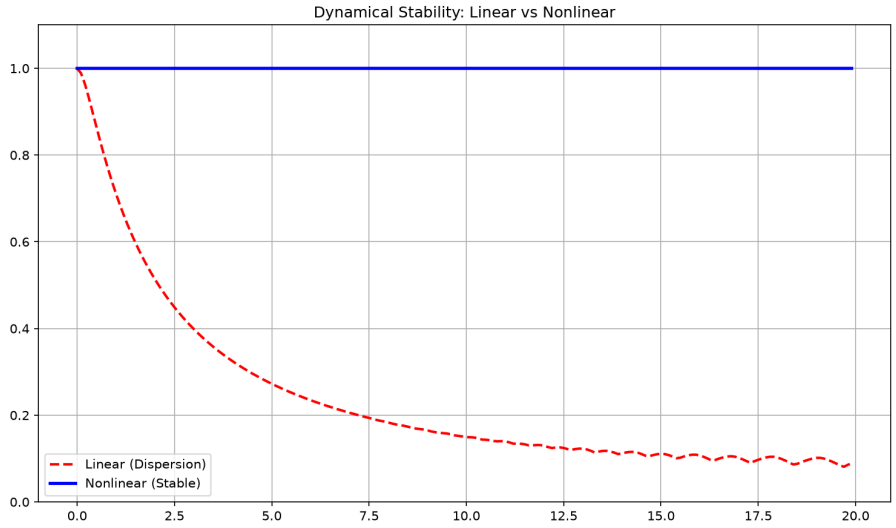


Figure 6: Dynamical stability of a 1D soliton. In a linear medium (red dashed), the wave disperses. In a nonlinear medium (blue solid), gravitational binding exactly balances dispersion, serving as a proof of concept for 3D topological stability.

5 Discussion and Conclusions

This work introduces a unified, reproducible computational synthesis connecting continuum elasticity, topological solitons, and analogue acoustic metrics. We demonstrate that gravitational-like attraction and lensing can be self-consistently modeled as mechanical strain relief in an elastic substrate.

Crucially, our dynamical simulations resolve the thermodynamic paradox. Gravity in this framework is not merely a passive byproduct of a static knot, but an active participant in its stability. The elastic restoring force creates an inward pressure that counteracts the dispersive tendency of the topological defect. This creates a self-regulating feedback loop (backreaction): the knot induces a strain field, and the resulting strain relief compresses the knot, stabilizing it against dispersion. This mechanical binding mechanism elegantly demonstrates that gravitational attraction is fundamentally a stabilizing, structure-forming force rather than a dispersive thermodynamic flow.

Declaration of Generative AI Use

The author utilized an AI language model for assistance in drafting, structuring, and refining the manuscript text, as well as optimizing the Python code for numerical integration. The author reviewed and edited all content, verified all mathematical equations and references, and takes full responsibility for the accuracy and integrity of the work.

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